# Agent-based Vision Mixer for Video Broadcasts

## Final Year Project - Mid Presentation

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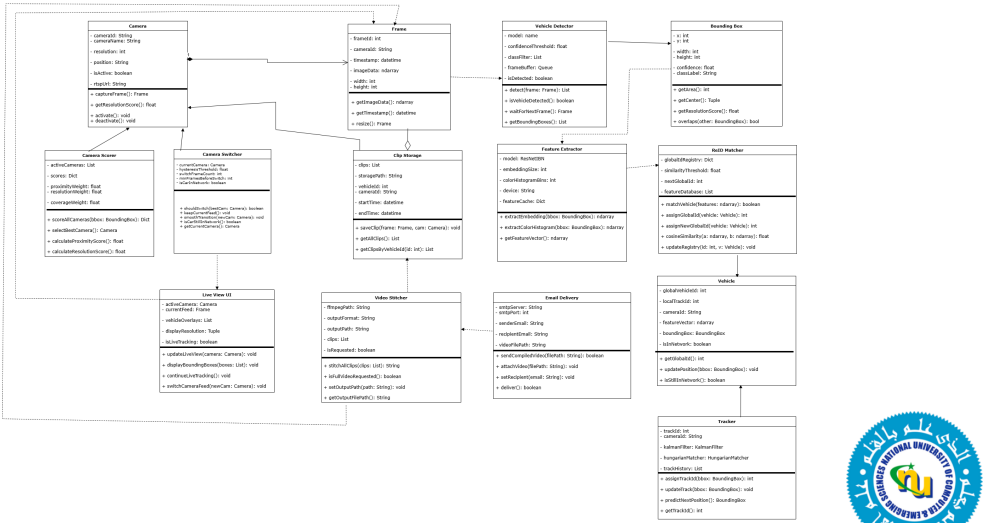


# Introduction

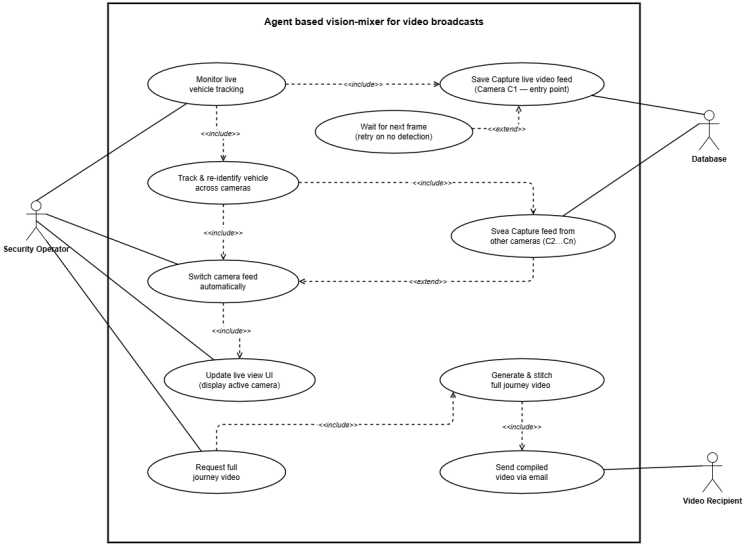
- Tracks vehicles across multiple cameras
- Use Models for detection and re-identification
- Automatically switches best camera view
- Generates summarized video of vehicle journey
- Sends video via email



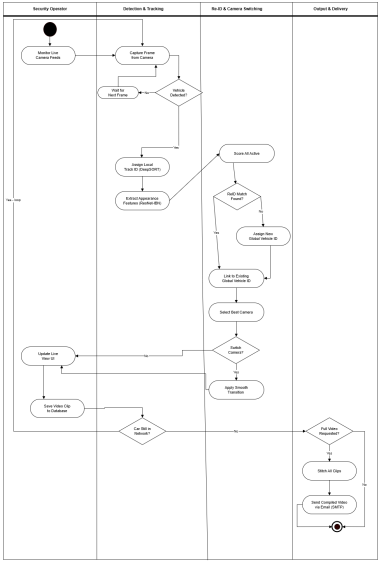
# Class Diagram



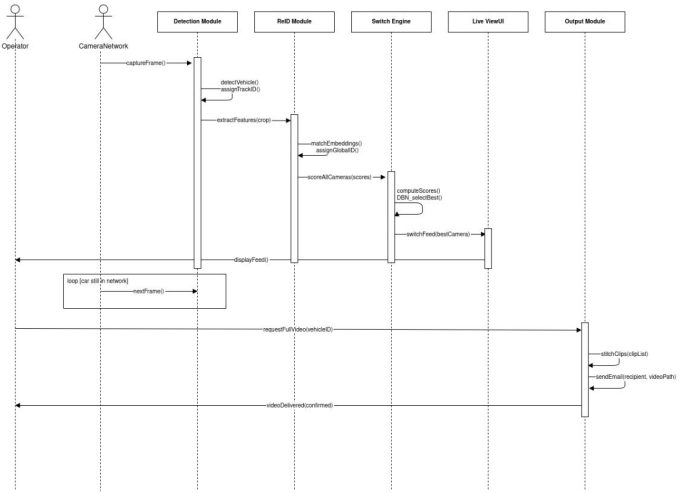
# Use Case Diagram



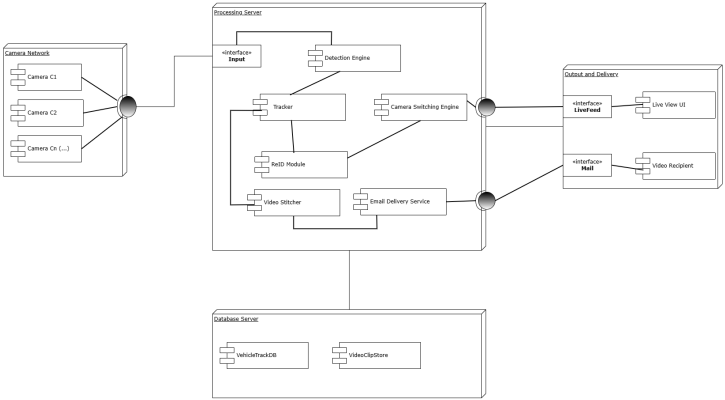
# Swimlane Diagram



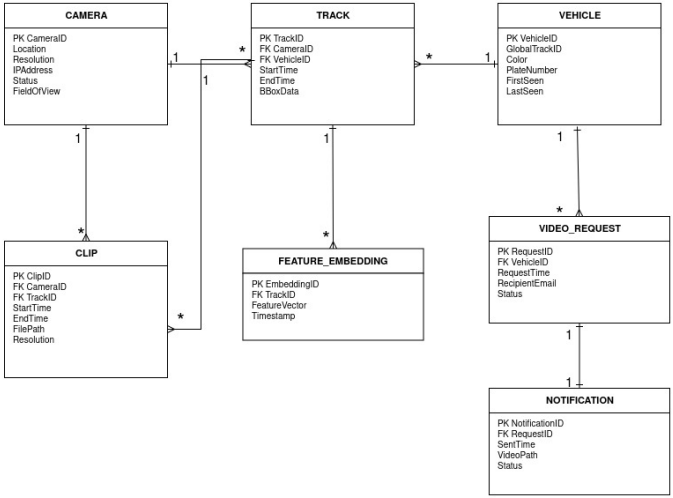
# Sequence Diagram



# Component Diagram



# ER Diagram


CAMERA
PK CameraID
Location
Resolution
IPAddress
Status
FieldOfView
TRACK
PK TrackID
FK CameraID
FK VehicleID
StartTime
EndTime
BBoxData
VEHICLE
PK VehicleID
GlobalTrackID
Color
PlateNumber
FirstSeen
LastSeen
CLIP
PK ClipID
FK CameraID
FK TrackID
StartTime
EndTime
FilePath
Resolution
FEATURE_EMBEDDING
PK EmbeddingID
FK TrackID
FeatureVector
Timestamp
VIDEO_REQUEST
PK RequestID
FK VehicleID
RequestTime
RecipientEmail
Status
NOTIFICATION
PK NotificationID
FK RequestID
SentTime
VideoPath
Status
1
*

# Test Cases

**Module 1: Vehicle Detection**

| Field | Description |
|---|---|
| Test Case ID | TC-VD-01 |
| Test Case Name | Detect Vehicle in Clear Frame |
| Module | Vehicle Detection |
| Objective | Verify that the model correctly detects a vehicle in a clear, unobstructed frame. |
| Preconditions | Model is loaded with COCO weights. Camera C1 is active and streaming. |
| Test Input | High-resolution frame containing a single car in daylight conditions. |
| Test Steps | 1. Start vehicle detection module.<br>2. Input frame from Camera C1.<br>3. Retrieve detections.<br>4. Verify class label and confidence score. |
| Expected Result | Detection with class = car and confidence 0.80. Bounding box displayed. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |

# Test Cases

**Module 1: Vehicle Detection**

| Field | Description |
|---|---|
| Test Case ID | TC-VD-02 |
| Test Case Name | No Detection on Empty Frame |
| Module | Vehicle Detection |
| Objective | Verify that the system returns zero detections when no vehicle is present. |
| Preconditions | Model is loaded. Camera feed is active. |
| Test Input | Frame showing empty road with no vehicles. |
| Test Steps | 1. Feed empty frame to detection module.<br>2. Retrieve detection list.<br>3. Check number of detections. |
| Expected Result | Detection list is empty. No bounding boxes. System waits for next frame. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |

# Test Cases

**Module 2: Camera Switching Engine**

| Field | Description |
|---|---|
| Test Case ID | TC-CS-01 |
| Test Case Name | Switch to Higher Resolution Camera |
| Module | Camera Switching Engine |
| Objective | Verify system switches to camera with higher resolution score. |
| Preconditions | Cameras C1 and C2 active. C2 has higher resolution. Same vehicle detected in both. |
| Test Input | Vehicle in C1 (0.55) and C2 (0.82). |
| Test Steps | 1. Provide scores to switching engine.<br>2. Call selectBestCamera().<br>3. Observe selected camera. |
| Expected Result | Camera C2 selected (higher score, above threshold). |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Test Cases

**Module 2: Camera Switching Engine**

| Field | Description |
|---|---|
| Test Case ID | TC-CS-04 |
| Test Case Name | Handle Overlap Zone Between Cameras |
| Module | Camera Switching Engine |
| Objective | Ensure best camera is selected in overlap and duplicate feed is avoided. |
| Preconditions | C1 and C2 detect same vehicle. Overlap detection enabled. |
| Test Input | C1 bbox = 2400 $px^2$, C2 bbox = 5100 $px^2$. |
| Test Steps | 1. Trigger handleOverlapZone().<br>2. Compare bbox area and resolution.<br>3. Verify active feed and duplication. |
| Expected Result | C2 selected (larger bbox). Single feed shown. No duplicate entry. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Test Cases

**Module 3: Re-Identification / Cross-Camera Matching**

| Field | Description |
|---|---|
| Test Case ID | TC-RI-01 |
| Test Case Name | Assign Global ID to New Vehicle |
| Module | ReID / Cross-Camera Matching |
| Objective | Verify new global ID is assigned when no match exists. |
| Preconditions | Feature database empty. ResNet-IBN model loaded. |
| Test Input | Cropped vehicle image from Camera C1 (first appearance). |
| Test Steps | 1. Extract feature embedding.<br>2. Call matchAcrossCameras().<br>3. Verify new global ID creation. |
| Expected Result | New ID (e.g., VH-001) created and stored with embedding. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Test Cases

**Module 3: Re-Identification / Cross-Camera Matching**

| Field | Description |
|---|---|
| Test Case ID | TC-RI-03 |
| Test Case Name | Reject Match for Different Vehicle |
| Module | ReID / Cross-Camera Matching |
| Objective | Ensure different vehicles are not assigned same ID. |
| Preconditions | VH-001 exists (red sedan). New blue truck appears. |
| Test Input | Cropped image of blue truck (Camera C2). |
| Test Steps | 1. Extract embedding.<br>2. Compute cosine similarity.<br>3. Check ID assignment. |
| Expected Result | Similarity $< 0.75$. New ID (VH-002) assigned. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Test Cases

**Module 4: Video Stitching and Email Delivery**

| Field | Description |
|---|---|
| Test Case ID | TC-VE-01 |
| Test Case Name | Stitch Clips in Chronological Order |
| Module | Video Stitching and Email Delivery |
| Objective | Verify clips are combined in correct time order. |
| Preconditions | Clips from C1, C2, C3 stored. FFmpeg installed. |
| Test Input | Clip A (0–15s), Clip B (14–28s), Clip C (27–40s). |
| Test Steps | 1. Add clips randomly.<br>2. Call stitch().<br>3. Verify output order. |
| Expected Result | Output follows A $\rightarrow$ B $\rightarrow$ C. Duration $\approx$ 40s. No errors. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Test Cases

**Module 4: Video Stitching and Email Delivery**

| Field | Description |
|---|---|
| Test Case ID | TC-VE-02 |
| Test Case Name | Deduplicate Overlapping Clip Segments |
| Module | Video Stitching and Email Delivery |
| Objective | Ensure overlapping clips keep only higher-quality segment. |
| Preconditions | C1 and C2 overlap. C2 has higher resolution. |
| Test Input | C1 (720p), C2 (1080p) same time window. |
| Test Steps | 1. Add both clips.<br>2. Call deduplicateOverlaps().<br>3. Stitch and verify output. |
| Expected Result | Only C2 clip retained. No duplicate time segments. |
| Actual Result | To be filled after execution |
| Pass / Fail | To be filled after execution |



# Literature Review - Paper Summaries (1/3)

## [1] Robust Deep Networks MO-MCT - *Multimedia Tools & Applications (2023)*

4-step deep learning pipeline: Mask R-CNN detection + Deep SORT tracking + ResNet-152 Re-ID with aggregation loss across 46 cameras. IDF1 = 0.8289 on AI City Challenge Track 3. Reuses ROI features to reduce training time.

## [2]Dynamic Global Tracker (DGT) - *Scientific Reports (2026)*

Online MCMT framework integrating cross-camera association directly into tracking. Hybrid Fusion Module (HFM) handles resolution disparity between cameras. Stable Trajectory Manager (STM) removes static/occluded noise. Achieves 90 FPS.

# Literature Review - Paper Summaries (2/3)

## [3] - Multi-Camera Vehicle Tracking - *CVPR Workshop (2022)*

2nd-place system at AI City Challenge 2022 (IDF1 = 0.8437). YOLOv5x6 detection + ResNet-IBN Re-ID + multi-level 4-round association + zone-based tracklet merging + spatial-temporal multi-camera matching with U-turn handling.

## [4] Multicamera Video Stitching Surveillance - *VAWKUM Trans.* (2019)

Combines multiple CCTV feeds into a single panoramic view. Pipeline: ORB feature identification → RANSAC matching → multi-band blending. SURF recommended for best performance. Temporal synchronization ensures frame-level alignment between streams.

# Literature Review - Paper Summaries (3/3)

## [5] SAE-MCVT Edge Computing Framework - *arXiv:2511.13904* (2025)

First scalable real-time MCVT on edge hardware (NVIDIA Jetson). YOLO + ByteTrack + ResNet-50 features on edge nodes; metadata aggregated at central server. Self-supervised KDE-based camera link model — no manual zone annotation required.

## [6] City-Scale MCVT Camera Link Model - *ResearchGate* (2025)

Improved self-supervised camera link model: clusters trajectory entry/exit positions and computes KDE-based transition time distributions automatically. Spatial-temporal constraints filter tracklet candidates before appearance matching - reduces false associations significantly.



# Literature Review - Feature Comparison

| Reference | Vehicle Detection | Re-ID | Camera Switching | Video Stitching | Real-Time Tracking | Auto Email |
|---|---|---|---|---|---|---|
| [1] Robust Deep Networks MO-MCT & Applications | ✓ | ✓ | – | – | – | – |
| [2] Dynamic Global Tracker (DGT) | ✓ | ✓ | – | – | ✓ | – |
| [3] Multi-Camera Vehicle Tracking | ✓ | ✓ | – | – | – | – |
| [4] Multicamera Video Stitching Surveillance | – | – | – | ✓ | – | – |
| [5] SAE-MCVT Edge Computing Framework | ✓ | ✓ | – | – | ✓ | – |
| [6] City-Scale MCVT Camera Link Model | ✓ | ✓ | – | – | – | – |
| **Our FYP** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

# Literature Review - Overall Conclusions (1/2)

## Detection is Solved

All 5 tracking papers use YOLO or Mask R-CNN. YOLOv8 is the current state-of-the-art baseline - we adopt this directly with COCO class filtering.

## Re-ID is the Core Challenge

ResNet-IBN with cosine similarity is the dominant approach for cross-camera matching - confirmed consistently across 4 of the 6 reviewed papers.

## Camera Switching = Our Novelty

**No reviewed paper** implements quality-based camera switching with hysteresis and overlap zone handling. This is our FYP's unique contribution.

# Literature Review - Overall Conclusions (2/2)

## Video Stitching is Understudied

Only Ali et al. (2019) addresses multi-camera video stitching. Our automatic clip assembly with timestamp deduplication fills a clear research gap.

## Real-Time is Achievable

DGT (90 FPS) and SAE-MCVT (IDF1 = 61.2 at 15 FPS on 2K streams) confirm that real-time MCMT is feasible — supporting our system's viability.

## Email Delivery is Unexplored

**No existing system** automates compiled journey video delivery via email. Our output module is entirely novel across all reviewed literature.



# Q & A

Thank You!

